

Raju Gudala

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About

Motivated Hardware Engineer with 1.5 years of hands-on experience in full-cycle hardware development from schematic capture and component selection through multilayer PCB layout, simulation, bring-up, and validation. Proficient in industry-standard design tools including KiCad and Altium Designer, with practical exposure to high-current power electronics, microcontroller-based embedded systems, and mixed-signal board design. Experienced in working across diverse hardware complexity levels from simple microcontroller breakout boards to complex multilayer, high-voltage power electronics designs. Quick to troubleshoot hardware issues and adapt to new domains, with a strong foundation in communication protocols, embedded firmware integration, and design-for-manufacturing (DFM) practices.

Experience

TiHAN-IIT Hyderabad

Research & Development Intern

May 2025 – Apr 2026

Full-time

Flight Controller | KiCAD, PCB Design

- Designed a custom UAV flight controller from scratch supporting ArduPilot firmware, routed on a 6-layer PCB with 250+ components centered around the STM32F427 as the primary flight MCU.
- Integrated STM32F103CBT6 as a dedicated I/O co-processor for PWM generation and peripheral management; interfaced critical IMU sensors over SPI/I²C and UART-based telemetry and GPS interfaces.
- Designed a dedicated onboard power supply system comprising multiple LDO and switching regulator stages to provide clean, regulated power rails to all flight-critical peripherals and sensor subsystems.

5V Power Module for UAV Avionics | KiCAD, LTspice, PCB Design

- Designed a high-efficiency 5V wide-input DC-DC power module supporting 3S–12S LiPo battery input using the LT8645SHV-2 synchronous buck converter.
- Integrated reverse-polarity protection, under-voltage lockout (UVLO), soft-start, and power-good monitoring to ensure reliable power delivery to flight-critical avionics.
- Optimized PCB layout for low EMI, low output ripple, and improved thermal performance, enabling stable operation of flight controllers, GPS, telemetry modules, and onboard sensors.

Custom Carrier Board for Cube Orange+ Flight Controller | KiCAD, PCB Design

- Designed a custom carrier board for the Cube Orange+ flight controller module, interfaced via board-to-board connectors, supporting full autopilot peripheral integration for UAV applications.
- Performed EMI/EMC-aware PCB layout with controlled impedance routing, proper ground plane management, and signal isolation between high-current and sensitive signal paths.
- Executed complete design flow — schematic capture, PCB layout, DFM review, and Gerber generation for fabrication; debugged and validated the fabricated board through bench testing and successful flight testing on a drone platform.

Tethered Drone Power System | Easy EDA, LTspice, Pspice, PCB Design

- Engineered end-to-end power delivery for a tethered drone platform, designed an AC-DC base station converter handling 230V AC input with 360V DC @ 8A output, and a high-voltage isolated DC-DC air unit converter stepping down 360V DC to 24V DC @ 100A (2.4kW) for avionics power distribution.
- Integrated voltage and current monitoring on the ground unit for real-time power telemetry; implemented automatic power source arbitration enabling seamless failover between the primary tether input and an onboard backup battery.
- Utilized TDK power conversion modules for high-reliability power stage implementation; validated system design using custom SPICE models in LTspice and executed full schematic capture and PCB design in EasyEDA.

Academic Projects

Digital Clock using 8051 | Keil μ Vision, Proteus Design Suite, Embedded C

- Developed a real-time digital clock on the 8051 microcontroller interfaced with a DS1307 RTC module.
- Implemented a bare-metal I²C driver from scratch in Embedded C to handle register-level communication between the MCU and RTC.
- Configured the DS1307 registers for timekeeping and retrieved real-time HH:MM:SS data for display on a 16x2 LCD.
- Simulated and validated the complete circuit in Proteus Design Suite prior to hardware implementation.

Adjustable DC-DC Step-Down Converter | KiCAD, LTspice, PCB Design

- Designed a wide-input adjustable DC-DC buck converter using the LM2596S-ADJ, accepting 5–40V input and delivering an adjustable output of 3.3–15V via a feedback potentiometer, rated for up to 3A continuous load current.

- Implemented input protection, EMI filtering, under-voltage lockout (UVLO), and a delay circuit to ensure stable input conditions before enabling the converter output.
- Designed the complete power conversion chain covering schematic capture, component selection, feedback network design, output filtering, and PCB layout for stable and efficient voltage regulation.

AgriSense : An Agricultural Bot | Arduino, IoT

- Designed an autonomous smart farm bot using Arduino UNO to collect real-time field data, monitor crop health, and measure soil moisture levels across multiple field locations.
- Implemented automated precision irrigation using a sprinkling motor triggered on low soil moisture readings, and integrated weed detection logic to cut weeds exceeding a defined height threshold.
- Optimized resource utilization and reduced manual labor by automating data collection and actuation tasks, enabling data-driven decision-making for sustainable farming practices.

Education

Rajiv Gandhi University of Knowledge Technologies
B.Tech - Electronics and Communication Engineering ; CGPA: 8.05

Nuzvid, Andhra Pradesh
2021-2025

Rajiv Gandhi University of Knowledge Technologies
Pre-University course (PUC); CGPA: 8.81

Nuzvid, Andhra Pradesh
2019-2021

Key Skills

PCB Design Tools: KiCAD, Altium Designer, Easy EDA, LTspice, PSpice
Programming Languages: C, Embedded C
Development Tools: STM32CubeIDE, Arduino IDE, Keil µVision, NI Multisim, Proteus
Microcontrollers: STM32 (F1/F4), ESP32, Arduino UNO, 8051, PIC, Raspberry Pi, 8086
Communication Protocols: UART, SPI, I²C, CAN, USB, WiFi
Soft Skills: Analytical Thinking, Team Collaboration, Time Management

Certifications

1. NPTEL Online Certification(SWAYAM MHRD)- Introduction to Industry 4.0 and Industrial Internet of things with 79%
2. NPTEL Online Certification(SWAYAM MHRD)- ESD:Hands-On Circuits and PCB Design with CAD Software with 78%
3. NPTEL Online Certification(SWAYAM MHRD)- VLSI Design Flow:RTL to GDS with 80%
4. NPTEL Online Certification(SWAYAM MHRD)- Introduction to Embedded Systems with 49%